Statistical Analysis Report

AI Salary Analysis DACH Region

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1. Overview

This report presents an exploratory statistical analysis of AI salary data across the DACH region (Germany, Austria and Switzerland). The analysis examines how compensation varies across experience levels, work modes, job titles and countries, using descriptive statistics, visualisations and hypothesis testing. The dataset is sourced from ai-jobs.net, a publicly available, crowdsourced salary platform. Download link: https://raw.githubusercontent.com/foorilla/ai-jobs-net-salaries/main/salaries.csv

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2. Dataset Description

The dataset is published by ai-jobs.net and contains anonymously submitted salary information from AI, ML and data science professionals worldwide. Submissions include job title, experience level, employment type, company size, company location, remote work ratio and annual salary in local currency, which the platform converts to USD using yearly average exchange rates sourced from the Bank for International Settlements.

For this project the dataset was filtered to DACH countries (DE, AT, CH), resulting in 875 rows. Two outlier entries were removed - one below the minimum plausible annual threshold of EUR 15,000 and one extreme value of EUR 753,480 - leaving a final working dataset of 873 rows across 11 columns. Key variables examined include salary_eur (derived by converting all salaries to EUR using live ECB exchange rates), experience_level (EN/MI/SE/EX), remote_ratio (0/50/100), company_location, and job_title.

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3. Methods

Descriptive Statistics

Summary statistics were computed using df.describe() to characterise the central tendency and spread of key numeric variables, particularly salary_eur. Categorical value counts were produced for experience_level, employment_type, company_size and company_location. This approach aligns with the initial data analysis (IDA) framework described by Lusa et al. (2024), who emphasise that systematic data screening - including examination of distributions, outliers and missing values - is an essential prerequisite to any inferential analysis and a cornerstone of reproducible research.

All salaries were converted to EUR using live exchange rates from the Frankfurter API (European Central Bank data) to enable direct cross-country comparison. This decision was made to avoid the distortion introduced by using USD as an intermediary currency for a regional European dataset.

Visualisations

Five visualisations were produced: (1) a salary distribution histogram by country, (2) a boxplot of salary by experience level, (3) a bar chart of average salary by top 10 job titles, (4) a boxplot of salary by work mode and (5) a correlation heatmap of numeric variables. Each visualisation was selected to address a distinct dimension of the data - distribution shape, group comparison and variable relationships. As recommended by Kass et al. (2016), visualising data before conducting formal tests is a best practice that reveals patterns and anomalies that summary statistics alone may obscure.

Hypothesis test

A one-way ANOVA was conducted to test whether work mode (on-site, hybrid, remote) has a statistically significant effect on salary. One-way ANOVA was appropriate because the dependent variable (salary_eur) is continuous and the independent variable is categorical with three groups (Emerson, 2022). The significance level was set at alpha = 0.05. Key assumptions include independence of observations, approximate normality within groups and homogeneity of variance. Given the large sample size of the on-site group (n=730) relative to hybrid (n=42) and remote (n=98), the assumption of equal variances should be interpreted with caution.

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4. Results

Descriptive statistics

After outlier removal, the mean DACH salary in EUR was approximately EUR 74,994 for on-site roles. Germany (DE) accounted for the largest share of entries, followed by Austria (AT) and Switzerland (CH). Austrian salaries clustered notably lower than German ones, while Swiss entries showed a wider spread with several values above EUR 150,000.

Visualisations

The salary distribution histogram (viz_descriptive_stats.png) confirmed right-skewed distributions for all three countries, consistent with typical labour market salary data. The experience-level boxplot showed a clear upward progression from Junior (EN) to Senior (SE), though the Executive (EX) group displayed a surprisingly flat median relative to Senior. The top-10 job titles bar chart (viz1_avg_salary_by_role.png) highlighted significant variation across roles. The work mode boxplot (viz2_salary_by_remote.png) showed a progressive increase from on-site to remote compensation. The correlation heatmap (viz3_correlation_heatmap.png) revealed weak correlations between salary_eur, remote_ratio and work_year.

Hypothesis test - Work mode

$F = 4.56$, $p = 0.0107$. The null hypothesis was rejected. Mean salary differs significantly across work modes, with remote roles averaging EUR 89,685 compared to EUR 74,994 for on-site roles and EUR 80,851 for hybrid.

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5. Interpretation

In plain terms, this analysis shows that AI professionals in Germany, Austria and Switzerland are not all paid the same - and the differences follow predictable patterns. More experienced professionals earn more, which is unsurprising, but the data also shows that where and how you work matters. Remote workers in the DACH region earn noticeably more on average than those working fully on-site. One likely explanation is that companies offering remote positions are competing for a wider, harder-to-find pool of talent and are willing to pay a premium to attract it. Switzerland stands out for having some of the highest individual salaries, though it contributes fewer entries to the dataset than Germany.

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6. Limitations and Potential Bias

Dataset limitation

The dataset is crowdsourced and relies on voluntary, anonymous submissions. This means it is not a representative sample of the DACH AI labour market.

Certain roles, industries or seniority levels may be systematically over- or underrepresented depending on which communities share and engage with the platform. Lusa et al. (2024) highlight that data collection processes and their potential biases must be examined and documented as part of any rigorous initial data analysis, as unexamined collection artefacts can propagate through all downstream conclusions.

Potential source of bias

The unequal group sizes in the work mode ANOVA (on-site n=730 vs. hybrid n=42 and remote n=98) introduce a risk of inflated variance estimates for smaller groups, which can affect the reliability of the F-statistic. Additionally, the dataset contains a mix of salary currencies including USD, EUR, CHF, GBP and INR, and while conversion to EUR was performed using current exchange rates, this does not account for the purchasing power differences or the year in which the salary was originally reported. Salaries reported in USD by DACH-based employers may reflect internationally benchmarked compensation rather than local market norms, potentially inflating the apparent salary level for certain roles.

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7. References

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